Exploration: Using Steric Parameters to Improve a Green Chemistry Reaction

An example of an organometallic reaction that makes a carbon-carbon bond with a high degree of stereoselectivity.[1] We will see how a group of researchers used steric parameters to investigate and improve the catalyst.

Grignard and Grignard-like reactions

You are familiar with the classic Grignard reaction. We make a metalated carbon nucleophile via oxidation of a metal, like zinc. We then add the nucleophilic carbon group to a solution containing an electrophile such as a ketone or aldehyde. The result is a new carbon-carbon bond.

The Grignard reaction requires dry conditions and aprotic solvents. The Barbier reaction is similar, but the zinc oxidation takes place *in situ* with the two carbon reactants. This reaction is insensitive to water and has been carried out in water solvent, making it a candidate for "green chemistry."[2]

The Nozaki-Hiyama-Kishi (NHK) reaction

The NHK reaction[3] uses chromium to enable the addition of an allyl bromide to a carbonyl group. The dirty little secret of all Grignard-like chemistry is that they are really organometallic reactions where the metal hides a radical mechanism. Chromium is not the greenest metal, but here it is being used in a catalytic amount rather than a stoichiometric amount.

A recent[4] example of a modified NHK reaction uses chiral ligands to provide stereoselectivity[5] for the addition reaction and features the use of steric LFER parameters to investigate the reaction.[1] By including a chiral ligand with the chromium salt, the authors achieved a high degree of stereoselectivity. Read this contribution and consider the following questions and discussion.

The catalytic cycle in figure 3 on the next page outlines a hypothesis for the mechanism of this version of the NHK reaction. The allyl bromide[6] is metalated via the oxidation of 2 eq. of Cr^{II} to Cr^{III}. The Cr^{III} is reduced by manganese metal to give the Mn^{II} salt byproduct and regenerate Cr^{II} so the cycle can continue. The trimethylsilyl chloride (TMSCl) is used to separate the alkoxide product from the Cr^{III} salt so that the salt is liberated to participate in the catalytic cycle. The initial silyl ether product is cleaved to give the alcohol during the workup procedure. TMS is the most sensitive of the silyl protecting groups.[7] The deprotection reaction is not described in the procedure,[1] but the conditions of the workup are what cleaves the silyl ether. It would have been nice if the authors had discussed this in the paper, but perhaps those familiar with the craft deem it obvious.

This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle*. Plots were typeset using *MatPlotLib* and *Python* tools. Diagrams were created and edited using *Affinity Designer*.

[1] "Examination of the Role of Taft-Type Steric Parameters in Asymmetric Catalysis." M.S. Sigman, J.J. Miller, *J. Org. Chem.*, **2009**, *74*, 7633-7643. https://doi.org/10.1021/jo901698t

Figure 1: A Grignard reaction.↓

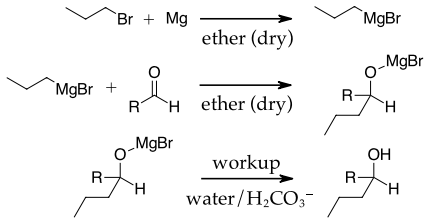
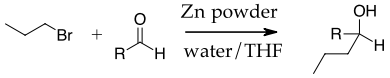


Figure 2: A Barbier reaction.↓



[2] Here is an example of a Barbier reaction performed using no solvent – very "green" indeed: "A Mechanochemical Zinc-Mediated Barbier-Type Allylation Reaction under Ball-Milling Conditions." J. Yin et al., *J. Org. Chem.*, **2020**, *85*, 2347-2354. https://doi.org/10.1021/acs.joc.9b02876

[3] For a useful dictionary of famous named reactions in chemistry see "Name reactions and reagents in organic synthesis" by B.P. Mundy, M.G. Ellerd and F.G. Favaloro, *J. Wiley & Sons, Hoboken, N.J.*, **2005**, ISBN 0-471-22854-0. It is available in the UPEI library: QD291.M86 2005.

[4] The word "recent" in physical organic chemistry means "this century."

[5] Chapter 6 of the textbook discusses stereochemistry in detail.

[6] I presented allyl chloride in figure 3 for simplicity. The allyl bromide mechanism is the same but you will have a mixture of bromide and chloride counterions.

[7] "Selective Deprotection of Silyl Ethers." T.D. Nelson, R.D. Crouch, *Synthesis*, **1996**, *1031-1069*. https://doi.org/10.1055/s-1996-4350 or "Greene's Protective Groups in Organic Synthesis" by P.G.M. Wuts, T.W. Greene, **2014**, *John Wiley & Sons, Inc.*, https://doi.org/10.1002/9781118905074. Available as an e-book via the UPEI library.

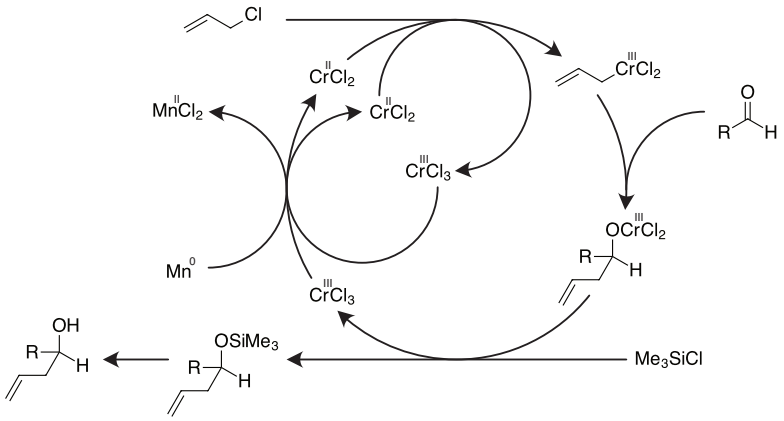


Figure 3: A mechanism for the Nozaki-Hiyama-Kishi reaction.
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Figure 4: A stereoselective Nozaki-Hiyama-Kishi reaction.[1]↓

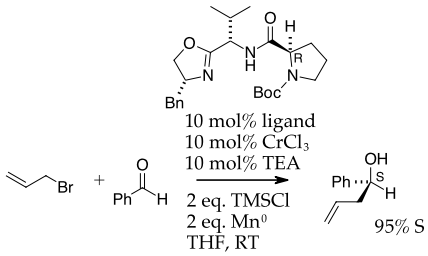
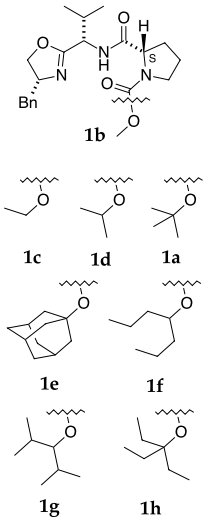


Figure 5: The ligand series explored in the paper.[1]↓



[8] "The upsilon steric parameter – definition and determination." M. Charton, *Topics Curr. Chem.*, **1983**, *114*, 57-91. https://doi.org/10.1007/BFb0111213

The Ligand Series

The ligand is chelated to the chromium centre and favours one enantiomeric product over the other during the addition reaction. The planar aldehyde group can approach the ligand/metalated carbon centre with either the *re* or *si* face. If we were to use the *R* ligand as shown in figure 5 then we have two diastereomeric transition states possible: the *R:re* and the *R:si* options. If the energy difference between the diastereomers is about 2.5 J mol^{-1} then that would give you 95 % of one enantiomeric product and 5 % of the other.

The authors chose a series of ligands with increasing bulk on the proline group and plotted the log of the enatiomeric ratio ($^R/_S$) vs. an LFER parameter for steric bulk.[1] They chose to use Charton parameters, υ, which are a calculated series based on the Van der Waals radius of the attached atoms.[8] I plan to plot the results against Taft parameters, E_s, in my own analysis of the results. Taft parameters are based on experimental observations, unlike Charton parameters that are based on mathematical calculations of atomic volumes. The authors of the paper make an argument that the Charton approach is superior, but I wonder if that is just because those parameters gave them better correlations.

						%R using this Electrophile[1]		
Ligand	Group	Substituent	E_s	E'_s [9]	υ_x [8]	Ph(CO)H	$Ph(CH_2)_2(CO)H$	$Ph(CO)CH_3$
1a	t-Bu	$C(CH_3)_3$	−1.54	−1.43	1.24	96	74	69
1b	Me	CH_3	0	0	0.52	60	51	25
1c	Et	C_2H_5	−0.07	−0.06	0.56	65.5	52.5	24
1d	i-Pr	$CH(CH_3)_2$	−0.47	−0.48	0.76	82.5	56.5	37.5
1e	1-adamantyl				1.33	96	73.5	72
1f	n-Pr_2CH	$CH(C_3H_7)_2$	−2.11	−2.03	1.54	73		37.5
1g	i-Pr_2CH	$CH(CH(CH_3)_2)_2$		−5.01	1.70	77.5		55
1h	Et_3C	$C(C_2H_5)_3$	−3.8	−5.29	2.38	91.5		61.5

Table 1: Experimental data reported in the paper.[1]↑

Data Analysis – Charton Plot

Now, let us do our own analysis of the data. We can now plot the $\log {}^{R}/_{S}$ vs. E_s (I will be using the improved Taft values, E'_s, developed by Dubois et al.)[9] and see if we get a similar correlation between enantiomeric ratio and size parameter as the authors.[10]

Before we do that, let us plot the data using the Charton steric parameter, υ, as the authors did in figure 5 of the paper[1] (see figure 6 of this document.) Take note of the argument that there are two regimes of steric sensitivity. The Taft plot has a slope of 1.5 for the substituents with less bulk, and then the stereoselectivity plummets and a new regime begins with a slope of 0.7.

I find this argument unsatisfying. The authors argue that steric strain favours one enantiomer over the other until a limit is reached where a new catalyst structure occurs, and then increased bulk again begins improving the stereoselectivity. There is clearly a trend of increasing stereoselectivity until a threshold is reached where the structures of the catalysts change in some manner. However, I do not find three-point plots very compelling. I favour the simpler hypothesis that increased steric bulk improves stereoselectivity until we go too far and it all goes to pieces. I do observe that, in the case of acetophenone, the authors did not propose a line fit for those three points.

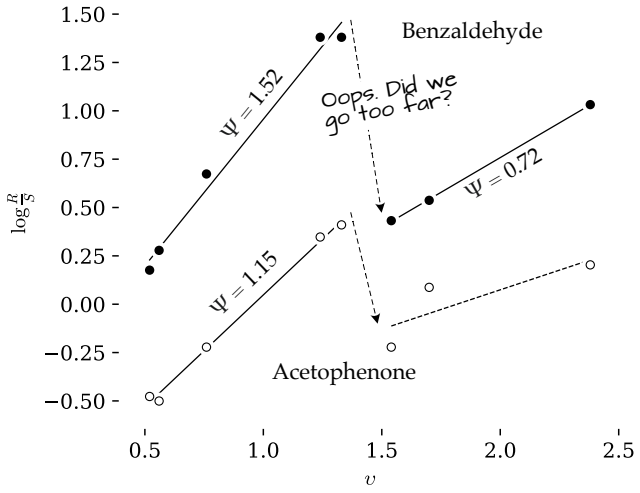


Figure 6: Plot of the log of the enantiomeric ratio vs. Charton steric parameters for the allylation of benzaldehyde and acetophenone as presented in figure 5 of the original paper.[1]
←
Experimental data is shown in table 1.

The *Python* notebooks used to create the plots in this document are available at https://github.com/blinkletter/4410PythonNotebooks/blob/main/Class_17/ChartonExploration/TaftExample.ipynb.

The images were then edited in *Affinity Designer* to add decorations such as labels and dotted lines.

The data set presented in table 1 is available at https://github.com/blinkletter/4410PythonNotebooks/blob/main/Class_17/ChartonExploration/Data2.csv

[9] "Steric effects–I: A critical examination of the taft steric parameter–E_s. Definition of a revised, broader and homogeneous scale. Extension to highly congested alkyl groups." J.A. MacPhee, A. Panaye, J.-E. Dubois, *Tetrahedron*, **1978**, *34*, 3553-3562. https://doi.org/10.1016/0040-4020(78)88431-2

[10] See the interactive Python notebook that accompanies this document for documentation of the calculations and plotting.

Data Analysis – Taft Plot

Charton steric parameters are designed to express the steric volume of the atoms closest to where the group is connected.[8] Taft steric parameters are based on experimental observations and provide a "real world" measurement of the steric effect of a group in acid-catalyzed ester hydrolysis.[9]. The two parameter sets usually track well together, but there seem to be exceptions.

In figure 7, I have plotted the same enantiomeric ratio data against Taft E'_s. The results are similar. The Taft E'_s list of parameters does not have a value for the adamantyl group and the values are more extreme for the bulkiest groups. Observe that the correlation in the second set of three points with the highest steric bulk is no longer telegraphing a straight line. The general conclusion remains the same – there is a direct correlation with steric parameters until a threshold is reached.

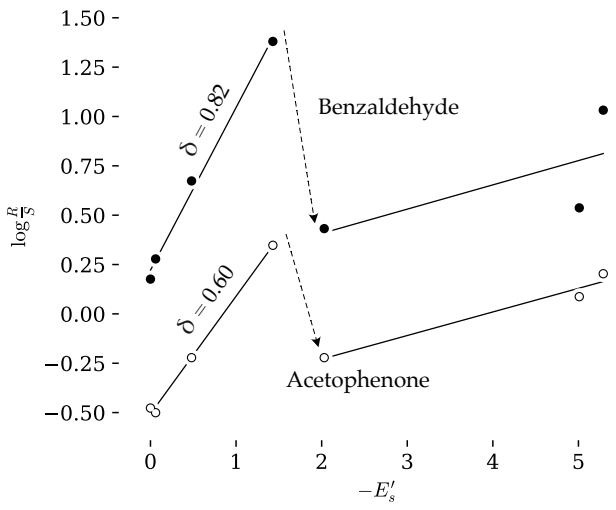


Figure 7: Plot of the log of the enantiomeric ratio vs. Taft steric parameters for the allylation of benzaldehyde and acetophenone. ⟵ Experimental data is shown in table 1.

Data Analysis – Comparing E'_s and υ

The authors of the paper were able to get a straighter line for the three-point plots after the break in the trend by using Charton υ parameters compared with Taft E'_s parameters. Perhaps this is why the author preferred Charton to Taft?

We can plot the data set of E'_s and υ against each other as shown in figure 8.[11] There are many substituents that have values for both parameters. The result is encouraging and disappointing. We see a clear and almost unshakeable correlation between the Taft E'_s and Charton υ parameters, with a few obvious exceptions. The calculated Charton parameters and the experimental Taft parameters match up almost perfectly. However, we see some deviations for very bulky substituents. Some bulky groups match the correlation while

[11] See the interactive Python notebook that accompanies this document for documentation of the calculations and plotting.

several not-too-dissimilar groups do not. Unfortunately for this study, two of the bulkiest groups used in the data fall far off the line.

The most extreme Taft steric parameters are part of the data set produced by Dublois et al.[9], and those authors did express their surprise at the high magnitude of the values. They did, however, assert their accuracy in their discussion. These values were obtained by competition experiments against other substituents, so there is the possibility of increased error. Perhaps they are just wrong?[12,13,14]

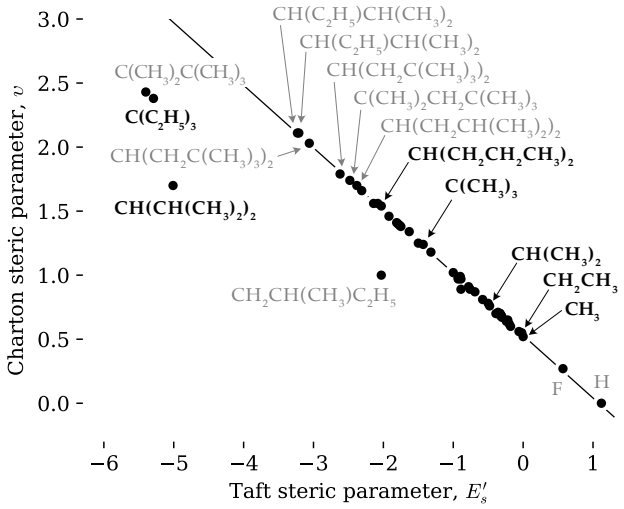


Figure 8: A plot of the correlation between the Taft E'_s and the Charton υ steric parameters.
←
The black labelled substituents are used in the plots here; the grey labelled substituents are shown for comparison.

One benefit of the Charton system is that one does not need to scour the literature looking for kinetic data of sterically-strained acid-catalyzed ester hydrolysis, adjust the values for temperature and reaction conditions and then use that experimental result to calculate a Taft E'_s value for that given group. A calculator may produce the parameter based on Van der Waals radii and Charton's mathematical recipe.

Conclusion

I began this exploration with a plan to show that the Taft E'_s values were just as good as the Charton υ parameters for analyzing the organometallic stereoselective reaction investigated by the authors. Ultimately, I have come full circle and now accept that the Charton values were chosen for a good reason. The authors did not trust the E'_s parameters for bulky substituents.[15]

One soothing fact is that groups for which the Taft and Charton values correlate include the first point after the break in the trend. So the enantioselectivity does increase with steric bulk, and then the catalyst "breaks" when the substituents are too bulky.

[12] "Steric effects-II: Relationship between topology and the steric parameter. E'_s – topology as a tool for the correlation and prediction of steric effects." A. Panaye, J.A. MacPhee, J.-E. Dubois, *Tetrahedron*, **1980**, *36*, 759-768, https://doi.org/10.1016/S0040-4020(01)93691-9

[13] "Steric effects—III: Composition of the E'_s parameter. Variation of alkyl steric effects with substitution. Role of conformation in determining sterically active and inactive sites." J.-E. Dubois, J.A. MacPhee, A. Panaye, *Tetrahedron*, **1980**, *36*, 919-928, https://doi.org/10.1016/0040-4020(80)80043-3

[14] "Steric effects. 4. Multiparameter correlation models. Geometrical and proximity site effects for carboxylic acid esterification and related reactions." J.A. MacPhee, A. Panaye, J.-E. Dubois, *J. Org. Chem.*, **1980**, *45*, 1164-1166. https://doi.org/10.1021/jo01294a052

[15] Also, the Charton method can be used to produce steric parameters for just about any group.

I still do not agree with declaring a correlation after the break without better information about the structural change. Just because the shattered remnants gave us three points that fell in a straight line does not guarantee it is a real trend.

Exploration Tips

This document is a good example of an exploration, albeit short. It features a plot of the data and a new plot using an alternate set of steric parameters. We were introduced to Charton parameters, a subject only briefly mentioned in class (if at all).

Exploration Questions

This exploration could easily be expanded by asking the following questions as you read the paper[1] and then writing a discussion that provides your answers.

1. Is the enantiomeric ratio, $^{R}/_{S}$, a measure of free energy difference and thus suitable for using in an LFER plot?
2. Why did the authors not include electronic effects in their consideration of substituent effects?
3. In the Charton plot, I observe a gap between $\upsilon = -1.33$ (adamantyl)and $\upsilon = -1.54$ ($CH(Pr)_2$). Somewhere in there may be a group that gives even better enantioselectivity before it goes to ruin. What substituents have an υ value of -1.45?[16]
4. Using your answer for the question above, propose a ligand that would potentially have a better $^{R}/_{S}$ ratio for product than ligand **1a**. Justify your choice of substituent. Now propose a synthesis of this ligand. Include all reaction conditions in your plan.[17]

[16] I have collected steric parameters in a file available from the course database at https://raw.githubusercontent.com/blinkletter/LFER-QSAR/refs/heads/main/data/Taft_Es_Williams.csv

[17] There might be a detailed description of the synthesis of **1a** out there somewhere that you could use as a guide. I wonder where one might find it?